

DAA UNIT WISE IMPORTANT QUESTIONS

UNIT-I

1. a) Distinguish between Algorithm and Pseudocode.
b) Define i) Profiling ii) Time Complexity iii) Space Complexity
c) Discuss the amortized analysis with an example.
2. a) Explain the properties / characteristics of an algorithm with an example.
b) Give the algorithm for matrix multiplication and find the time complexity of the algorithm using step count method.
c) Explain recursive functions algorithm analysis with an example.
3. a) Explain the Asymptotic notations with example.
b) Prove $3n^3 + 2n^2 = O(n^3)$; $3^n \neq O(2^n)$.
4. a) How the performance can be analyzed? Explain with the example.
b) Give the algorithm for transpose of a matrix $m \times n$ and find the time complexity of the algorithm using step count method.

UNIT-II

1. a) Explain partition exchange sort (Quick Sort (or) in-place Sort (or) unstable sort) algorithm and trace this algorithm for $n=8$ elements: 24,12, 35, 23,45,34,20,48.
b) Write Divide – And – Conquer recursive Quick sort algorithm and analyze the algorithm for average time complexity.
c) Derive the time complexity of Quick sort algorithm for worst case.
2. a) Write the General method and Control Abstraction of Divide – and – Conquer.
b) What is stable sorting method? Is Merge sort a stable sorting method?
c) What are the advantages and disadvantages of Divide – And – Conquer?
3. a) Distinguish between Merge sort and quick sort.
b) Explain Recursive Binary search algorithm with suitable examples.
c) Discuss the time complexity of Binary search algorithm for best and worst case.
4. a) Explain Merge sort (Stable sort (or) Not in-place) algorithm and trace this algorithm for $n=8$ elements: 24,12, 35, 23,45,34,20,48
b) Derive the time complexity of Merge sort algorithm for all cases.

UNIT-III

1. a).Write the general method and Control Abstraction of Greedy method
b) Define i) Principles of optimality ii) Feasible solution iii) Optimal solution.
c) Distinguish between Prim's and Kruskal's spanning tree algorithm.
2. a) State the Greedy Knapsack? Write the algorithm for Greedy knapsack and also compute the time complexity.
b) Find an optimal solution to the knapsack instance $n=7$ objects and the capacity of knapsack $m=15$. The profits and weights of the objects are $(P_1, P_2, P_3, P_4, P_5, P_6, P_7) = (10, 5, 15, 7, 6, 18, 3)$ $(W_1, W_2, W_3, W_4, W_5, W_6, W_7) = (2, 3, 5, 7, 1, 4, 1)$.
3. a) State the Job – Sequencing Deadline Problem
b) Find an optimal sequence to the $n=5$ Jobs where profits $(P_1, P_2, P_3, P_4, P_5) = (20, 15, 10, 5, 1)$ and deadlines $(d_1, d_2, d_3, d_4, d_5) = (2, 2, 1, 3, 3)$.
4. a) What is a Spanning tree? Explain Prim's Minimum cost spanning tree algorithm with suitable example and also find the time complexity.
b) What is a Minimum Cost Spanning tree? Explain Kruskal's Minimum cost spanning tree algorithm with suitable example and also find the time complexity.
5. a) Discuss the single – source shortest paths (i.e. Dijkstra's) algorithm with suitable example and also find the time complexity.

UNIT-IV

1. a) Distinguish between Dynamic Programming and Greedy method.
b) Explain how Matrix – chain Multiplication problem can be solved using dynamic programming with suitable example.
2. a) Draw all possible binary search trees for the identifier set (do, if, stop).
b) Draw an Optimal Binary Search Tree for $n=4$ identifiers $(a_1, a_2, a_3, a_4) = (\text{do}, \text{if}, \text{read}, \text{while})$ $P(1:4) = (3, 3, 1, 1)$ and $Q(0:4) = (2, 3, 1, 1, 1)$.
3. Describe the Dynamic 0/1 Knapsack Problem. Find an optimal solution for the dynamic programming 0/1 knapsack instance for $n=3$, $m=6$, profits are $(p_1, p_2, p_3) = (1, 2, 5)$, weights are $(w_1, w_2, w_3) = (2, 3, 4)$.
4. a) What is All – Pair Shortest Path problem (APSP)? Discuss the Floyd's APSP algorithm and discuss the analysis of this algorithm.
b) What is a principle of optimality? Explain how travelling sales person problem uses the dynamic programming technique with example and also find space and time complexity.
5. Give the statement of Reliability design problem and explain with suitable example.

UNIT-V

1. a) What is a Backtracking and give the 4 – Queens's solution. Draw the portion of the state space tree for $n = 4$ queens using backtracking algorithm.
b) Write an algorithm for N – queen's problem. Give time and space complexity for 8 – queen's problem.
2. a) Define: i) State Space tree ii) E – Node and iiii) Dead Node.
b) Give the statement of sum –of subsets problem. Find all sum of subsets for $n=4$, $(w_1, w_2, w_3, w_4) = (11, 13, 24, 7)$ and $M=31$. Draw the portion of the state space tree using fixed – tuple sized approach.
3. a) Define Chromatic number & Give the state space tree for 4 – Coloring problem.
b) Explain the Graph – coloring problem. And draw the state space tree for $m= 3$ colors $n=4$ vertices graph. Discuss the time and space complexity.
4. a) Write the Control Abstraction of iterative Backtracking method
b) What is a Hamiltonian Cycle? Explain how to find Hamiltonian path and cycle using backtracking algorithm.

UNIT-VI

1. a) Distinguish between fixed – tuple sized and variable tuple sized state space tree organization.
b) Define: i) LC – Search ii) Branch and Bound (BB) iii) FIFO – BB.
c) Write the Control Abstraction of Least – Cost Branch and Bound.
d) Distinguish between backtracking and branch – and bound techniques.
2. a) Define Bounding Function? Give the statement of 0/1 Knapsack FIFO BB and explain the procedure with the knapsack instance for $n=4, m=15, (p_1, p_2, p_3, p_4) = (10, 10, 12, 18)$ $(w_1, w_2, w_3, w_4) = (2, 4, 6, 9)$.
b) Give the 0/1 Knapsack LCBB algorithm. Explain how to find optimal solution using variable – tuple sized approach.
3. a) Explain travelling sales person problem LCBB procedure with the following instance and draw the portion of the state space tree and find an optimal tour.

$$\begin{pmatrix} \infty & 20 & 30 & 10 & 11 \\ 15 & \infty & 16 & 4 & 2 \\ 3 & 5 & \infty & 2 & 4 \\ 19 & 6 & 18 & \infty & 3 \\ 16 & 4 & 7 & 16 & \infty \end{pmatrix}$$

- b) Draw the portion of state space tree generated by LCBB for the 0/1 Knapsack instance: $n = 5$, $(p_1, p_2, \dots, p_5) = (10, 15, 6, 8, 4)$, $(w_1, w_2, \dots, w_5) = (4, 6, 3, 4, 2)$ and $m=12$. Find an optimal solution using fixed – tuple sized approach.
4. a) Draw the portion of state space tree generated by FIFOBB for the job sequencing with deadlines instance $n=5$, $(p_1, p_2, \dots, p_5) =(6, 3, 4, 8, 5)$, $(t_1, t_2, \dots, t_5) = (2, 1, 2, 1, 1)$ and $(d_1, d_2, \dots, d_5)=(3, 1, 4, 2, 4)$. What is the penalty corresponding to an optimal solution.